TASK 6

a) **Develop a pig Latin script to count the number of words in a text file.**

1. nano input.txt

   Hadoop is powerful
   Pig is easy to use
   Hadoop and Pig work together
   Pig is powerful and flexible
2. hdfs dfs -mkdir -p t6/
3. hdfs dfs -put input.txt t6
4. nano wordcount.pig

```
lines = LOAD 't6/input.txt' USING TextLoader() AS (line:CHARARRAY);

words = FOREACH lines GENERATE FLATTEN(TOKENIZE(line)) AS word;

filtered_words = FILTER words BY word IS NOT NULL;

grouped_words = GROUP filtered_words BY word;

word_count = FOREACH grouped_words GENERATE group AS word, COUNT(filtered_words) AS count;

STORE word_count INTO 't6/output' USING PigStorage(',');
```

5. pig -x mapreduce wordcount.pig

b) **Develop a pig Latin script to find maximum temperature across all the years using weather data set.**

1. nano weather.txt

   01/01/2010, 5.2, 15.8
   05/02/2010, 8.5, 18.4
   10/03/2010, 10.1, 22.6
   15/04/2010, 12.3, 25.9
   20/05/2010, 15.0, 30.5
   25/06/2011, 18.2, 32.4
   30/07/2011, 20.8, 35.7
   05/08/2011, 22.1, 36.2
   10/09/2011, 16.4, 28.5
   15/10/2011, 12.9, 24.3
   20/11/2012, 6.5, 20.7
   25/12/2012, 3.2, 18.6
   01/01/2013, 2.8, 15.9
   10/02/2013, 5.7, 19.3
   20/03/2013, 7.4, 21.8

30/04/2013, 9.6, 23.5
15/06/2013, 17.3, 38.2
10/07/2013, 21.7, 40.6
05/08/2014, 23.1, 41.3
10/09/2014, 20.9, 39.8

2. hdfs dfs -put weather.txt t6
3. nano max_temperature.pig

   weather_data = LOAD 't7/weather.txt' USING PigStorage(',') AS (date:CHARARRAY, min_temp:FLOAT, max_temp:FLOAT);

   max_temperature = ORDER weather_data BY max_temp DESC;

   highest_temp = LIMIT max_temperature 1;

   final_output = FOREACH highest_temp GENERATE date, max_temp;

   STORE final_output INTO 't7/output2' USING PigStorage(',');

4. pig -x mapreduce max_temperature.pig

Output:

```
[cloudera@quickstart Desktop]$ pig -x mapreduce max_temperature.pig
```

```
cloudera@quickstart:~/Desktop
File  Edit  View  Search  Terminal  Help
140)
        at org.apache.pig.tools.grunt.GruntParser.parseStopOnError(GruntParser.j
ava:202)
        at org.apache.pig.tools.grunt.GruntParser.parseStopOnError(GruntParser.j
ava:173)
        at org.apache.pig.tools.grunt.Grunt.exec(Grunt.java:84)
        at org.apache.pig.Main.run(Main.java:613)
        at org.apache.pig.Main.main(Main.java:158)
        at sun.reflect.NativeMethodAccessorImpl.invoke0(Native Method)
        at sun.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.
java:57)
        at sun.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAcces
sorImpl.java:43)
        at java.lang.reflect.Method.invoke(Method.java:606)
        at org.apache.hadoop.util.RunJar.run(RunJar.java:221)
        at org.apache.hadoop.util.RunJar.main(RunJar.java:136)
2025-03-19 00:47:25,102 [main] INFO  org.apache.hadoop.mapred.ClientServiceDeleg
ate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirect
ing to job history server
2025-03-19 00:47:25,457 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.mapReduceLayer.MapReduceLauncher - Success!
```

```
ne.mapReduceLayer.MapReduceLauncher - Success!
[cloudera@quickstart Desktop]$ hdfs dfs -cat bbc/output2/*
05/08/2014,41.3
```